

Practical 1: Introduction to React and Component Architecture

Objective	To set up a React development environment using Vite and build a static UI using independently structured, reusable components.
Problem Statement	• Create a React application using Vite for a student portfolio page. • The application must include at least 4 reusable components: Header, About, Skills, and Footer. • Each component must be independently structured and composed into a single-page layout. • No logic or JSX should be duplicated across components. • Props must be used to pass at least one piece of data into a minimum of 2 components.
Project Structure	<pre> student-portfolio/ ├── src/ │ ├── App.jsx │ └── components/ │ ├── main.jsx │ ├── Header.jsx │ ├── About.jsx │ ├── Skills.jsx │ └── Footer.jsx </pre>
Implementation	<pre> D:\>node -v v24.11.1 D:\>npm -v 11.6.2 </pre> To Create folder with react and JavaScript write the following command: <code>npm create vite@latest student-portfolio -- --template react</code> <code>npm install</code> <code>npm run dev</code> <pre> PROBLEMS DEBUG CONSOLE CODE REFERENCE LOG OUTPUT PORTS TERMINAL PS D:\CollegeWork\5th Sem\AWDF\Practical 1> npm run dev > practical_1@0.0.0 dev > vite Port 5173 is in use, trying another one... VITE v8.1.3 ready in 490 ms → Local: http://localhost:5174/ → Network: use --host to expose → press h + enter to show help </pre>

Header.jsx:

```
function Header({ name, role }) {  
  return (  
    <header>  
      <h1>{name}'s Portfolio</h1>  
      <p>{role}</p>  
    </header>  
  )  
}  
  
export default Header
```

About.jsx:

```
function About({ bio, education }) {  
  return (  
    <section>  
      <h2>About Me</h2>  
      <p>{bio}</p>  
      <p>Education: {education}</p>  
    </section>  
  )  
}  
  
export default About
```

Skills.jsx:

```
function Skills({ skills }) {  
  return (  
    <section>  
      <h2>Skills</h2>  
      <ul>  
        {skills.map((skill, index) => (  
          <li key={index}>{skill}</li>  
        ))}  
      </ul>  
    </section>  
  )  
}  
  
export default Skills
```

Footer.jsx:

```
function Footer({ name, year }) {  
  return (  
    <footer>  
      <p>© {year} Jemit One page portfolio. All rights reserved.</p>  
    <div>
```

```
        <a href="https://github.com/Vaghasiya-Jemit-kanaiyalal" target="_blank"
rel="noopener noreferrer">Github</a>
        <a href="https://www.linkedin.com/in/jemitvaghasiya/" target="_blank"
rel="noopener noreferrer">LinkedIn</a>
        <a href="https://jemitportfolio.netlify.app/" target="_blank"
rel="noopener noreferrer">Portfolio</a>
    </div>
</footer>
)
}

export default Footer
```

App.jsx:

```
import Header from './components/Header'
import About from './components/About'
import Skills from './components/Skills'
import Footer from './components/Footer'

function App() {
  const portfolio = {
    name: 'Jemit Vaghasiya',
    role: 'Full Stack AI Developer',
    bio: 'I'm Jemit Vaghasiya, a Computer Science Engineering student passionate
about full-stack web development, AI, and open-source contributions.',
    education: 'B.Tech in Computer Science, Graduation: 2028',
    skills: ['HTML5', 'CSS3', 'JavaScript', 'React', 'Vite', 'Git', 'node.js',
'Express.js', 'Data Analusis', 'Machine Learning', 'Python', 'C++'],
  }

  const currentYear = new Date().getFullYear()

  return (
    <div>
      <Header name={portfolio.name} role={portfolio.role} />
      <main>
        <About bio={portfolio.bio} education={portfolio.education} />
        <Skills skills={portfolio.skills} />
      </main>
      <Footer name={portfolio.name} year={currentYear} />
    </div>
  )
}

export default App
```

ESLint (bundled with the Vite React template) was run to check for unused variables, bad practices, and syntax issues:

```
PS D:\CollegeWork\5th Sem\AWDF\Practical 1> npm run lint

> practical_1@0.0.0 lint
> eslint .
```

Output

Jemit Vaghasiya's Portfolio

Full Stack AI Developer

About Me

I'm Jemit Vaghasiya, a Computer Science Engineering student passionate about full-stack web development, AI, and open-source contributions.

Education: B.Tech in Computer Science, Graduation: 2028

Skills

HTML5 CSS3 JavaScript React Vite Git node.js Express.js Data Analysis
Machine Learning Python C++

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[Github](#) [LinkedIn](#) [Portfolio](#)

GitHub Repo along with commands:

```
git init
```

```
git remote add origin
```

```
git add .
```

```
git commit -m "Basic portfolio with Header, About, Skills, Footer components"
```

```
git branch -M main
```

```
git push -u origin main
```

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Files main Go to file

Practical 1

- public
- src
- .gitignore
- README.md
- eslint.config.js
- index.html
- package-lock.json
- package.json
- vite.config.js

Lab_Work / Practical 1

Vaghasiya-Jemit-kanaiyal Basic portfolio with Header, About, Skills, Footer components ed37fb3 · 1 minute ago History

Name	Last commit message	Last commit date
..		
public	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
src	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
.gitignore	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
README.md	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
eslint.config.js	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
index.html	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
package-lock.json	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
package.json	Basic portfolio with Header, About, Skills, Footer components	1 minute ago
vite.config.js	Basic portfolio with Header, About, Skills, Footer components	1 minute ago

README.md

React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](#) uses [Oxc](#)
- [@vitejs/plugin-react-swc](#) uses [SWC](#)

React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](#).

Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](#) for information on how to integrate TypeScript and [typescript-eslint](#) in your project.

Conclusion

The React project was created using Vite, and a simple student portfolio page was developed. Four reusable components (Header, About, Skills, and Footer) were used to build the page. Props were used to pass data like the student's name and skills to different components, making the code simple, reusable, and well organized.